Systems Engineering – Lab 4

Task 1: Quality Metric for the E-Conference Project

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1. Short Context of the Project

Our team of four students is working on the E-Conferen make it harder for authors, reviewers and organizers to work smoothly with the system. ce system, a web-based platform that supports the "Emerging Software Technologies" conference. The main goal of the system is to automate and manage the key business processes of a scientific conference: user registration, creation and maintenance of user profiles, paper submission, automatic allocation of reviewers, review collection, and the final acceptance or rejection decision for each paper.

Technically, the project uses a Java Spring Boot back-end, a web front-end built with HTML, CSS and JavaScript, and a relational database. The system is intended to serve many conferences and journals, handle thousands of users, and provide secure and reliable access for authors, reviewers and organizers.

2. Proposed Metric

Name of the metric: Paper Submission Success Rate (PSSR).

The Paper Submission Success Rate measures how many paper submission attempts are completed successfully, without system errors or the user giving up, compared to the total number of submission attempts in a given time interval. In simple terms, it answers the question: "Out of all the times authors tried to submit a paper, in how many cases did everything work correctly and the paper reached the system in a valid state on the first try?".

For our project this metric is especially relevant because several causes from the fishbone diagram are directly related to the submission process: incomplete author submissions, missing fields in user profiles, inconsistent metadata, bugs in the registration module and database connection errors. Whenever these problems occur, they will usually lead to failed or repeated submission attempts and therefore reduce the PSSR value.

3. Why This Metric Is Important

For the E-Conference system, the paper submission workflow is one of the most critical features. If authors cannot easily and reliably submit their papers, the conference cannot function. A low submission success rate usually indicates usability problems (confusing forms, unclear validation messages, missing profile fields) or reliability problems (registration bugs, database connection errors, timeouts).

Our analysis of the system shows that inefficiency comes from both human and technical causes. On the human side, we see issues such as limited organizer experience, weak feedback to authors, manual reviewer assignment and poor deadline management. On the technical side, there are bugs and connection errors. By tracking the Paper Submission Success Rate, we get a simple measure of whether these problems are under control from the authors' perspective. If the PSSR increases after we fix defects or improve the feedback in the user interface, we can be confident that our changes are having a positive effect.

Because our project is not a long individual thesis but a group project with four people, we need a quality indicator that is easy to compute and easy to explain to stakeholders. PSSR satisfies this: if the percentage is high, most authors manage to submit their papers successfully; if the percentage drops, we clearly have a quality issue that we must investigate using the causes already mapped in the fishbone diagram.

4. Measurement Function (Formula)

We define the metric over a chosen time period (for example, one week of conference submissions):

PSSR = (Number of successful first-attempt submissions / Total number of submission attempts) × 100%

where:

- A submission attempt is counted every time an author starts the submission process and presses the final "Submit" button.
- A successful first-attempt submission is an attempt where the paper is stored in the system, assigned an ID, and confirmed to the user, without server errors and without the user needing to restart the process from the beginning.

In practice, problems identified in the fishbone diagram (for example bugs in the registration module, database connection errors or incomplete author submissions) will increase the number of failed attempts in the denominator and therefore lower the PSSR value.

5. Scenario for Collecting the Data

Imagine that the conference opens the submission site for two weeks. During this period, the system logs every paper submission attempt with the following information: timestamp, user ID, paper ID (if created), and status of the attempt ("success", "validation error", "server error", or "abandoned"). It also logs technical events such as database connection errors and application exceptions from the registration module.

At the end of the two weeks, we export the log and count:

- The total number of submission attempts.
- The number of attempts that ended with status "success" on the first try.

For example, suppose there were 200 submission attempts. Out of these, 170 were successful on the first try, 20 failed because of validation errors or missing profile fields that forced the authors to restart the process, and 10 ended with server errors caused by bugs or database connection problems. Then:

PSSR = (170 / 200) × 100% = 85%

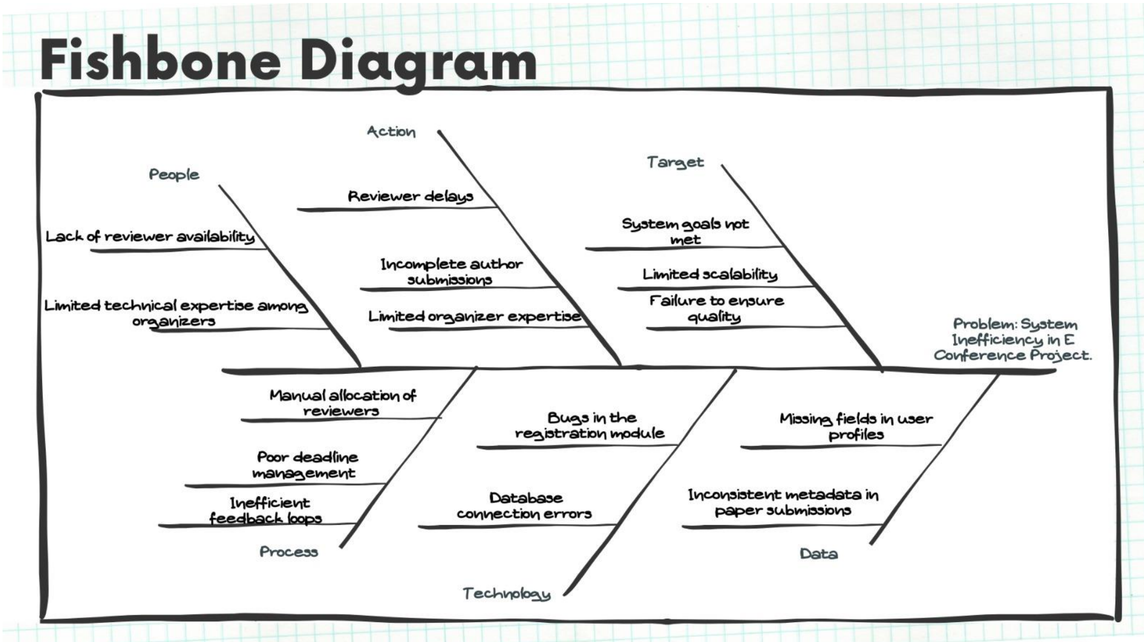
Based on this value, our team can set a quality goal, for example: "During the real conference, the Paper Submission Success Rate should be at least 95%." If we see that the metric is lower than the target, we can use the categories from the fishbone diagram (People, Process, Technology, Data, Action, Target) to systematically search for root causes and decide what to improve next.

6. Quality Attribute and Quality Model

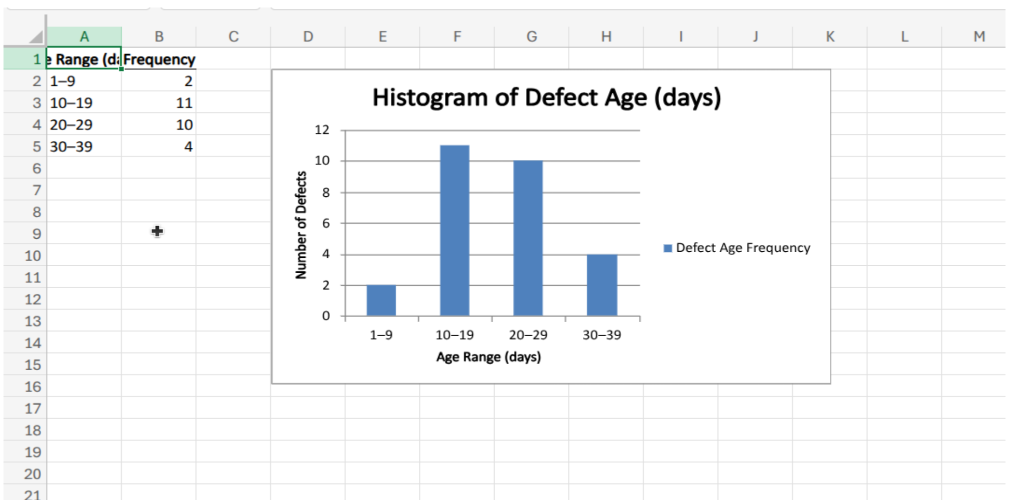
The Paper Submission Success Rate mainly reflects how usable the system is. When the interface is clear, error messages are easy to understand, required fields in profiles and submissions are well designed, and the workflow is intuitive, authors are able to submit their papers correctly on the first attempt. At the same time, the metric is sensitive to reliability problems, because crashes, bugs in the registration module or database connection errors will immediately lower the success rate.

Overall, PSSR is mapped to McCall's classic quality model, where it contributes primarily to the Usability factor while also capturing aspects of Reliability inside the same model. By monitoring this metric over time, we obtain a concrete, measurable indicator of how well the E-Conference system supports users in carrying out one of its key tasks—submitting papers.

7. Ishikawa Diagram



8. Histogram



Conclusions from the histogram:

- Most defects take between 10 and 29 days to be resolved.
- Very few defects are resolved in under 10 days (only 2).
- A smaller number of defects take over 30 days to be resolved (4 defects).
- The typical defect age is roughly in the middle bins (10–29 days), which suggests that the process tends to resolve issues within about two to four weeks.